ASTANA IT UNIVERSITY

DEVELOPMENT OF A WEB PLATFORM FOR MANAGING THE FACULTY MENTORSHIP SYSTEM IN A HIGHER EDUCATION INSTITUTION

Diploma Project Pre-Defense

Student: Tulegenova Karina
Group: SE-2419
Supervisor: Tulebayev Yersultan

ASTANA IT UNIVERSITY

PROJECT PROPOSAL

1

Problem statement

Fragmented manual processes for mentor requests, selection.

2

Aim

Develop a web platform to manage mentoring and support teachers' professional growth.

3

Objectives

Digitize mentor assignments, individual plans, development visits and progress reviews.

4

Target users

AITU mentors and mentees, School directors and UMO coordinators.

5

Expected outcomes

A working platform with clear progress tracking, centralized records and mentoring reports.

6

Project scope

AITU mentoring lifecycle: assignment to final review, with role-based access.

EVIDENCE OF RELEVANCE

Roles and confidentiality	01	At the meeting held on October 8, clients identified roles for mentees, mentors, School management and UMO. The platform should separate access and keep observation records confidential.
Support requests and assignment	02	Mentees describe their professional support needs before mentor assignment. Digital request forms and mentor profiles should support the School's decision.
Development plans and feedback	03	Mentoring follows an individual development plan. The platform should record progress checks, lesson observations and feedback between the mentor and mentee.
Monitoring and reporting	04	UMO and School management need an overview of mentoring activity. Reports should summarize meetings, recommendations and checklist results, with an export option.

Clients: Shunkeyeva Saule Alisherovna – Head of the Educational and Methodological Office, Abisheva Sandugash Kanatovna – Chief Manager of the Educational and Methodological Office, Tulebayev Yersultan Bakhytovich – Senior Lecturer and Mentor

LITERATURE REVIEW

Mentoring foundations	Digital support	Peer observation	Feedback and reflection
[1] Nuis et al. (2023) Conceptualizing mentoring in higher education: A systematic literature review Finding: define clear goals and roles.	[4] Bell (2024) Affordances, constraints and collaborative potential of e-mentoring platforms and their features for K–12 educators: A systematic literature review Finding: access needs sustained engagement.	[7] Cutroni & Paladino (2023) Peer-ing in: A systematic review and framework of peer review of teaching in higher education Finding: document the observation cycle.	[10] Cano García (2024) Peer feedback for teaching professional development: Conditions for it to take effect Finding: trust and feedback training matter.
[2] Abdollahi & Heshmati Nabavi (2023) Mentoring as an appropriate strategy for medical faculty member development in higher education: A systematic review Finding: evaluate the full mentoring cycle.	[5] Jin et al. (2024) Technological affordances in teachers' online professional learning communities: A systematic review Finding: support knowledge sharing.	[8] Otero-Saborido et al. (2024) Peer observation of teaching in higher education: Systematic review of observation tools Finding: include evidence and feedback fields.	[11] Hunt et al. (2021) Automated feedback is nice and human presence makes it better: Teachers' perceptions of feedback by means of an e-portfolio enhanced with learning analytics Finding: retain human discussion.
[3] López-Aguado & Scialdone (2026) La mentoría como estrategia para la mejora docente en la universidad, una revisión sistemática Finding: support formal mentoring pairs.	[6] Huang et al. (2024) A systematic review of technology-enabled teacher professional development during COVID-19 pandemic Finding: connect tasks with reflection.	[9] De la Iglesia et al. (2024) Peer feedback in teacher professional development: A systematic review Finding: guide useful follow-up actions.	[12] Wang et al. (2024) Effects of the visual analytics of peer feedback on teachers' TPACK development in a lesson study Finding: link summaries to original feedback.

Table 1. Literature review [1–12]. Findings span different contexts and require validation at AITU.

LITERATURE SYNTHESIS

Focus and sources	Evidence synthesis	Boundary of evidence	AITU design / evaluation
Evidence base [1–3, 7]	Mentoring and peer review need explicit goals, defined roles and a structured development process.	Published models span different populations and institutional settings.	Pair assignment, individual plans, meeting records and a complete observation cycle.
Design priorities [4–6, 8–12]	Digital tools support access and reflection when people provide useful feedback and sustained support.	Activity counts and positive perceptions alone do not demonstrate teaching improvement.	Private feedback, concise forms, reminders and interpretable progress summaries.
Project contribution Synthesis of [1–12]	The reviewed literature informs support practices and evaluation of the proposed AITU workflow.	This review does not establish that no comparable platform exists. Local effectiveness remains untested.	Evaluate task completion, record completeness, usability and feedback usefulness in an AITU pilot.

Table 5. Literature synthesis and design implications.

Research gap: Existing research examines mentoring, peer observation and feedback, but provides limited evidence on how these practices can be integrated into a single workflow for managing faculty mentorship in a university.

EXISTING SYSTEM COMPARISON

01

T O G E T H E R

Skill-based matching, goals, meetings and programme reports [13].

02

C H R O N U S

AI-assisted matching, guided meetings and progress dashboards [14, 19, 20].

03

I R I S C O N N E C T

Lesson videos and time-linked feedback [15]. Development goals and progress dashboards [16, 17].

SYSTEM COMPARISON MATRIX

Criterion	Together [13]	Chronus [14,19,20]	IRIS [15–18]
Mentor matching	✓	✓	✗
Mentor profiles / preferences	✓	✓	✗
Goals / development plans	✓	✓	✓
Meeting / session support	✓	✓	✓
Lesson observation	✗	✗	✓
Feedback	✓	✓	✓
AITU plan approval workflow	✗	✗	✗
Reporting / progress dashboards	✓	✓	✓

AITU needs a unified workflow for requests, mentor assignments, approved plans, feedback and archiving.

Table 6. Comparison of existing mentoring systems.

TECHNOLOGY COMPARISON

Decision	Alternative 1	Alternative 2	Alternative 3	Selected option
Backend	Django + DRF: auth, admin, ORM; conventions	FastAPI: typed API, OpenAPI; assemble admin	NestJS: modular TypeScript; separate Python ML	Django + DRF: workflow focus
Database	PostgreSQL: relations, JSONB; migrations	MySQL: relational storage; verify features	MongoDB: flexible documents; model relations	PostgreSQL: relational records
Communication	REST: resource endpoints; extra requests	GraphQL: client-selected data; resolver complexity	WebSocket: live two-way channel; state handling	REST: initial CRUD workflow

Table 7. Technology alternatives and selected options.

Functional Requirements: Mentee Portal

ID	Requirement
FR-MEN-01	Authentication using a corporate account or a test account for the prototype.
FR-MEN-02	View and update personal profile information where permitted.
FR-MEN-03	Submit a mentorship request specifying professional goals and development needs.
FR-MEN-04	Browse the mentor directory and indicate a preferred candidate.
FR-MEN-05	View the status of the mentorship request and mentor assignment.
FR-MEN-06	Collaborate on an individual development plan, save drafts, and submit the plan for approval.
FR-MEN-07	View the decision on the plan and comments when it is returned for revision.
FR-MEN-08	Create and confirm meeting records.
FR-MEN-09	Record interim feedback and adjustments to the development plan.
FR-MEN-10	Submit a request to change or terminate the mentorship program early.
FR-MEN-11	Complete the final checklist and self-assessment.
FR-MEN-12	Receive notifications and view the current status of the mentorship program.

Table 8. Functional requirements: mentee portal.

Functional Requirements: Mentor Portal

ID	Requirement
FR-MTR-01	Create and update the mentor profile, including experience, School, areas of expertise, request types, and interaction formats.
FR-MTR-02	View the verification and publication status of the mentor profile.
FR-MTR-03	View assigned mentorship requests and mentor–mentee pairs within the scope of authorized permissions.
FR-MTR-04	Confirm availability for mentorship and participate in the assignment approval process.
FR-MTR-05	Collaborate on the individual development plan and manage its versions.
FR-MTR-06	Create and confirm meeting records.
FR-MTR-07	Record brief meeting outcomes and link them to specific activities in the development plan.
FR-MTR-08	Provide interim feedback and propose adjustments to the development plan.
FR-MTR-09	Submit a request to modify or terminate the mentorship program.
FR-MTR-10	Prepare a final assessment report.
FR-MTR-11	View the current active mentorship workload.

Table 9. Functional requirements: mentor portal.

Functional Requirements: UMO / Programme Administrator

ID	Requirement
FR-UMO-01	Coordinate university-wide mentoring programmes within authorized responsibilities.
FR-UMO-02	Manage reference data and programme settings through the administrative interface.
FR-UMO-03	Check the completeness of documents and mandatory forms.
FR-UMO-04	Monitor statuses, deadlines, mentor workload, and indicators of inactivity.
FR-UMO-05	Generate analytics, reports, and data exports for a selected period.
FR-UMO-06	Access the archive of completed programmes and log views and exports.
FR-UMO-07	Review requests to correct documents after programme completion.
FR-UMO-08	Access authorized assessment data without automatic access to confidential consultations.
FR-UMO-09	Search and filter by full name, School, request area, status, deadlines, and completeness.

Table 10. Functional requirements: UMO administration.

Functional Requirements: Individual Development Plan

ID	Requirement
FR-PLAN-01	The mentor and mentee jointly complete the electronic plan.
FR-PLAN-02	Each activity specifies its area, content, deadline, interaction format, and expected outcome.
FR-PLAN-03	Create the plan within 10 working days of appointment and submit it for approval.
FR-PLAN-04	The School Director approves the plan or returns it with a mandatory comment.
FR-PLAN-05	Store the history of versions, comments, and decisions.
FR-PLAN-06	Restoring an earlier version creates a new version without deleting subsequent changes.
FR-PLAN-07	Record the completion status and a brief outcome for each activity.
FR-PLAN-08	Revise the plan while preserving its history.

Table 11. Individual development plan requirements.

Functional Requirements: Technical Administrator

ID	Requirement
FR-TECH-01	Maintain the application, technical accounts, service integrations, and logs.
FR-TECH-02	Configure SSO, email notifications, storage, and backups where authorized.
FR-TECH-03	View technical errors, integration statuses, and notification delivery statuses.
FR-TECH-04	Have no automatic access to confidential consultation content or substantive programme data.
FR-TECH-05	Log actions that may affect personal data or the archive.

Table 12. Technical administrator requirements.

NON-FUNCTIONAL REQUIREMENTS

Responsive design	01	Responsive interface for desktop, tablet, and mobile; no separate mobile app is required.
Multilingual interface	02	Interface available in Russian, Kazakh, and English.
Selective translation	03	User entries and substantive documents are not translated automatically; only the interface, system reference data, and messages are translated.
Corporate identity	04	Consistent visual style aligned with AITU's corporate identity.
Browser compatibility	05	Support for modern browsers and correct rendering of Kazakh characters.
Accessibility and usability	06	Basic accessibility and clear error messages.
Data integrity	07	Draft autosaving and protection against concurrent editing conflicts.
Performance	08	Target response time: up to 2 seconds for core operations and up to 3 seconds for search and filtering under the agreed workload.
Modular architecture	09	Modular, maintainable, and portable architecture.

ACCEPTANCE CRITERIA

1. Users can only access features and data permitted by their role and participation in the program.
2. Mentees can create and submit mentorship requests.
3. Mentees can browse the catalog and specify a preferred mentor.
4. Authorized users can record an agreed mentor assignment.
5. Mentors and mentees can jointly develop a plan and submit it for approval.
6. The School Director can approve the plan or return it for revision with a comment.
7. The system retains plan versions and a history of changes.
8. Mentors and mentees can create and confirm meeting records.
9. Confidential materials are inaccessible to users without special authorization.
10. Requests to modify the program can be submitted and processed.
11. Once all mandatory final forms have been completed, the system closes and archives the program.
12. The Academic Affairs Office (UMO) can view the status dashboard and export data within its access permissions.
13. All significant changes, access to sensitive data, and data exports are recorded in the Audit Log.
14. XLSX imports detect errors and duplicates before data is loaded.
15. The system retains completed programs and historical versions of documents.

SYSTEM CONTEXT DIAGRAM

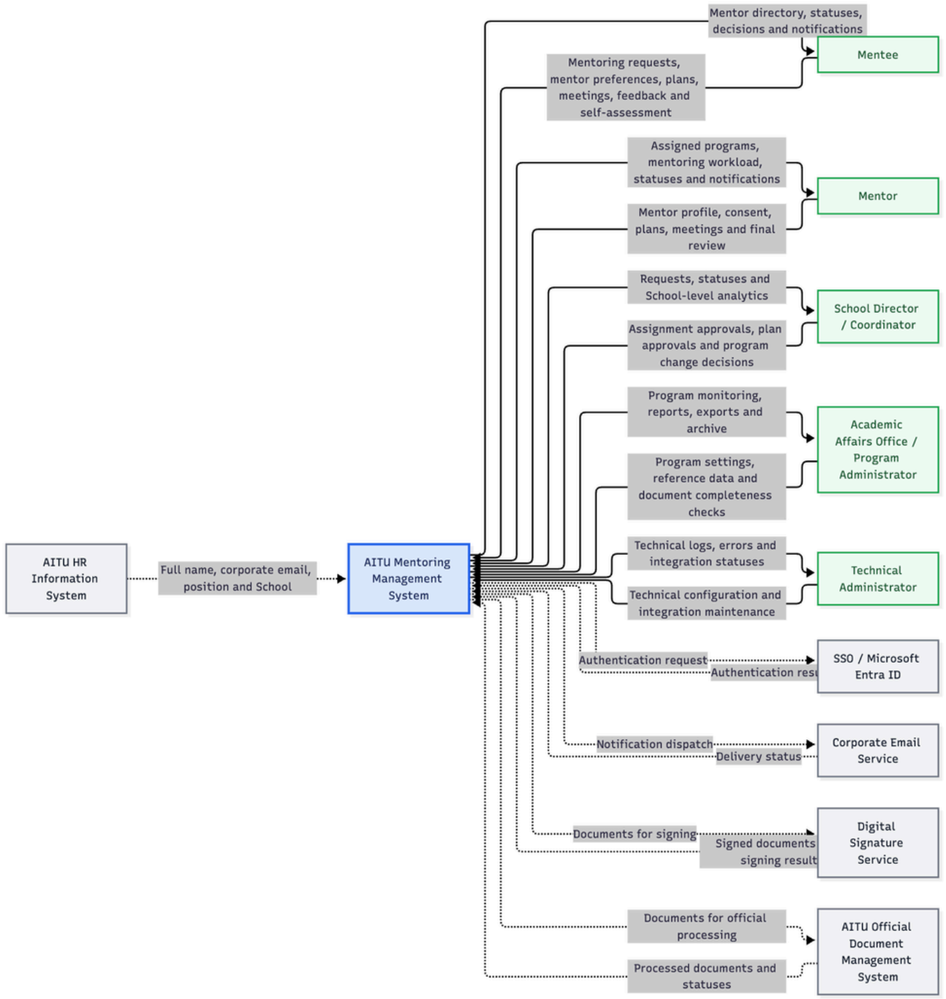


Figure 1. System context diagram.

USE CASE DIAGRAM

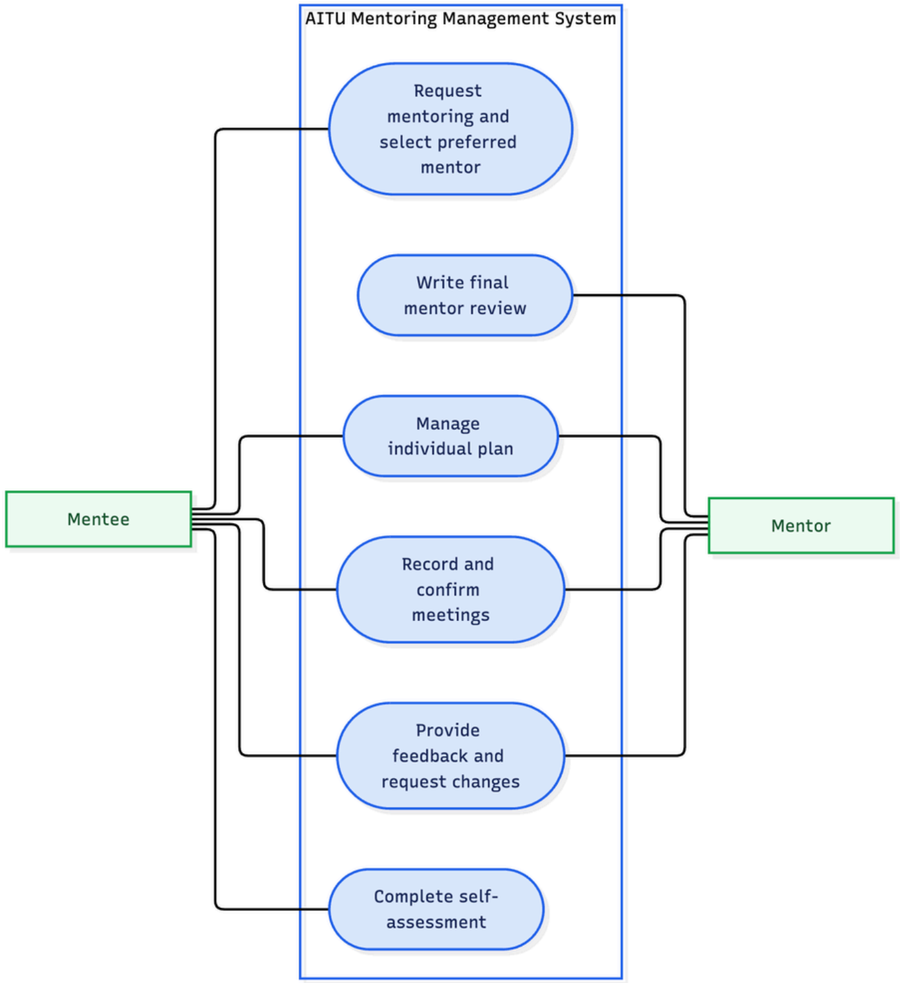


Figure 2. Use case diagram.

COMPONENT DIAGRAM

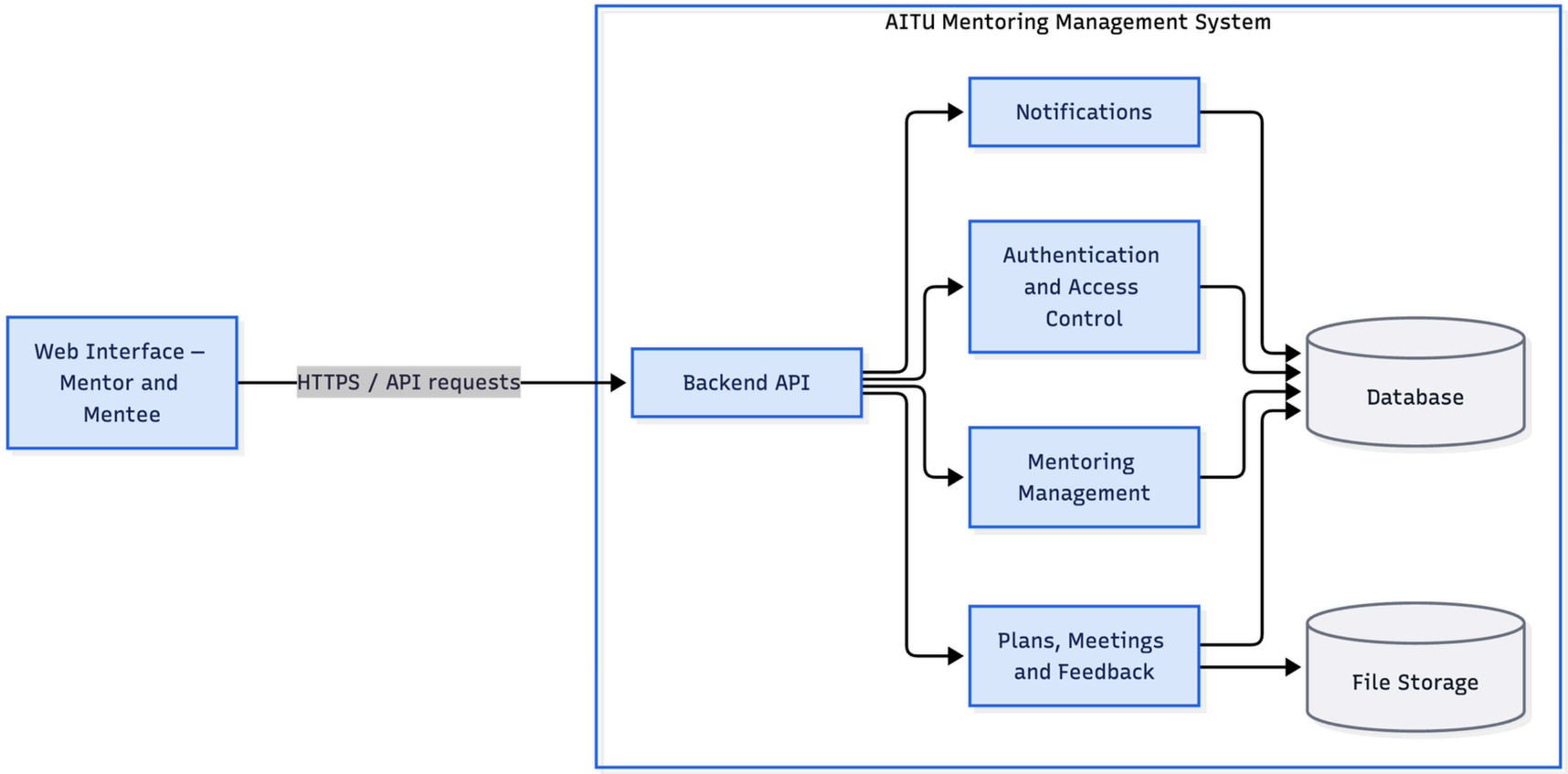


Figure 3. Component diagram (initial draft).

UX / PROTOTYPE

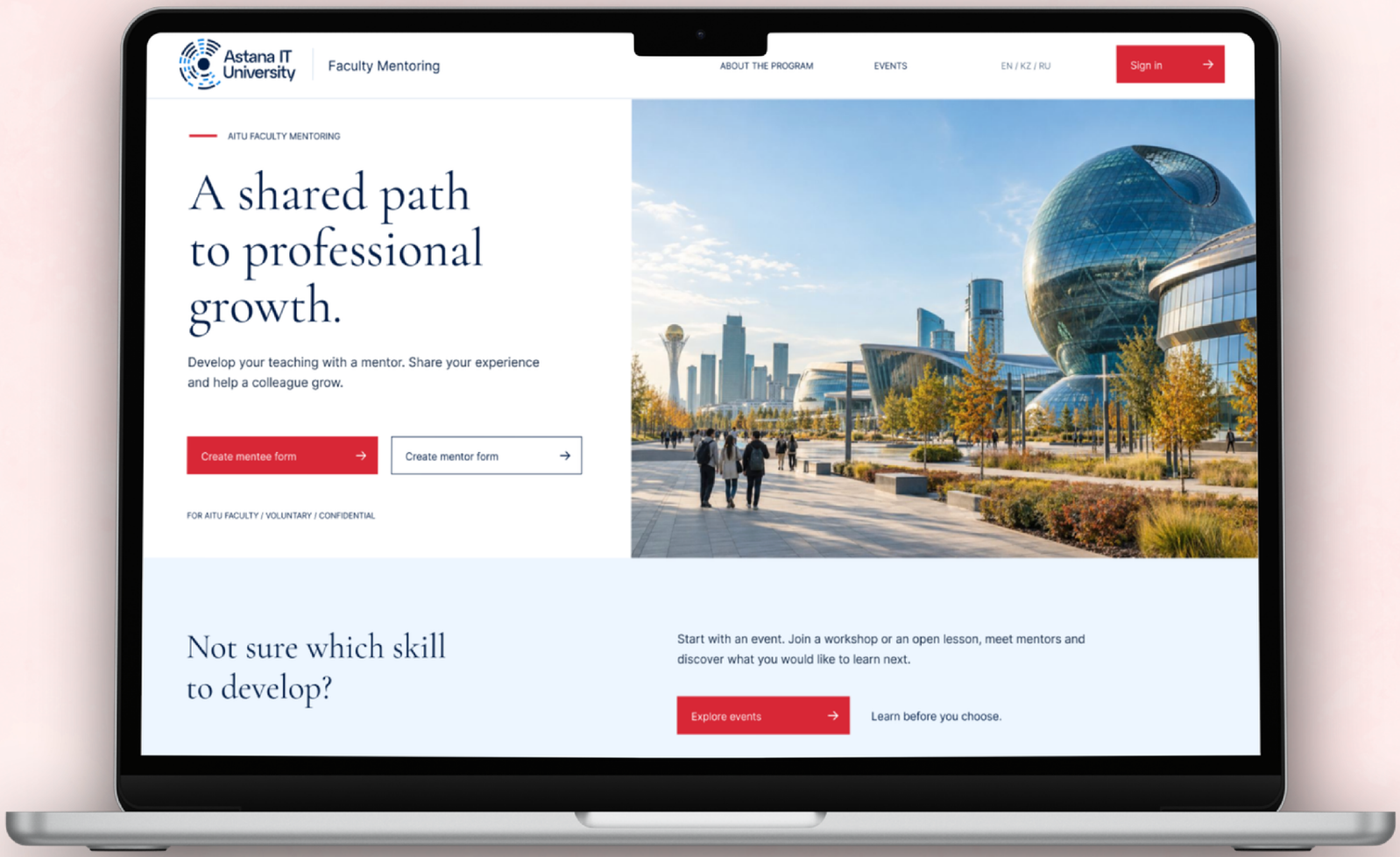


Figure 4. Landing page and mentoring entry points.

MENTEE DASHBOARD

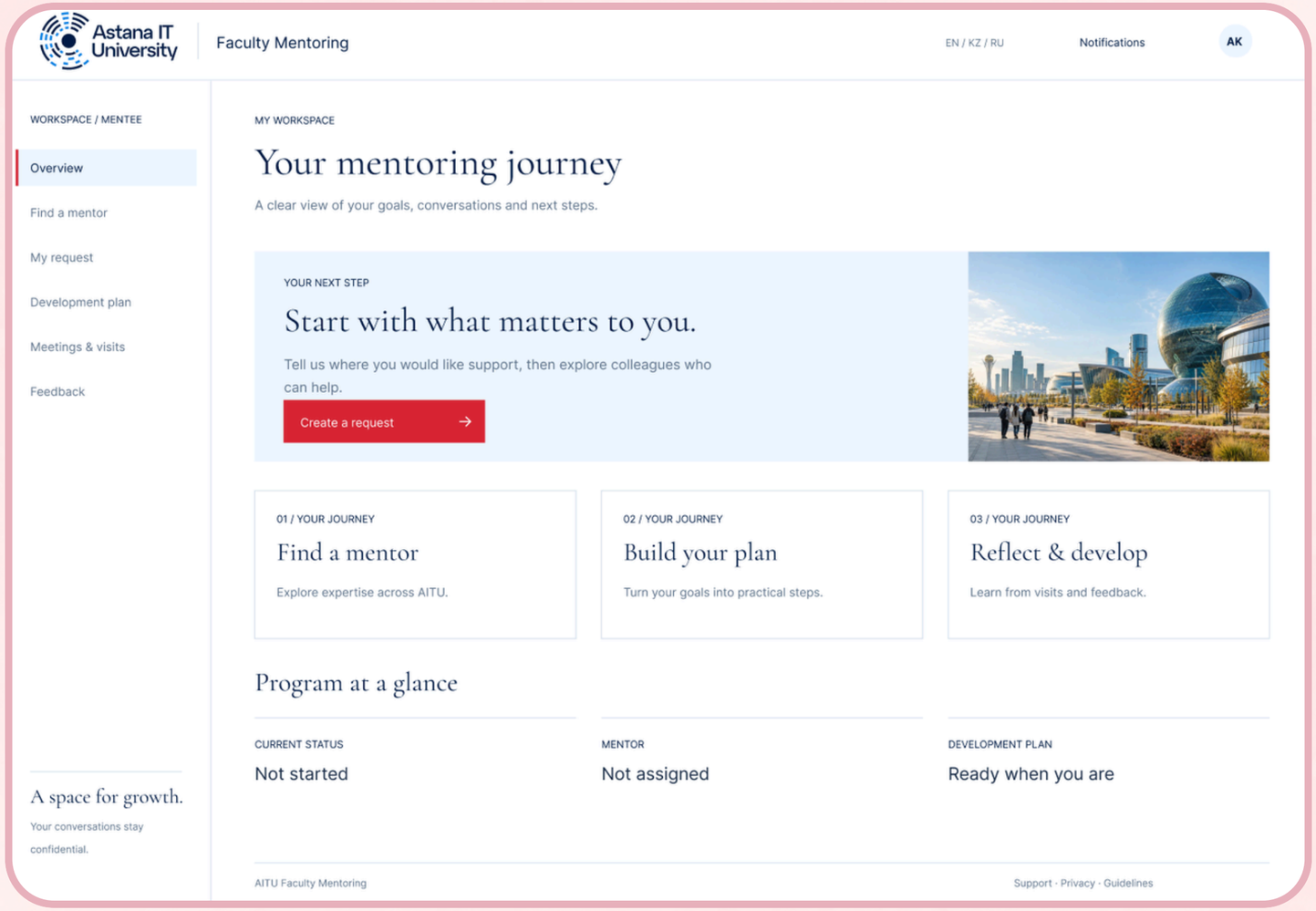


Figure 5. Mentee dashboard and mentoring journey.

MENTOR CATALOGUE

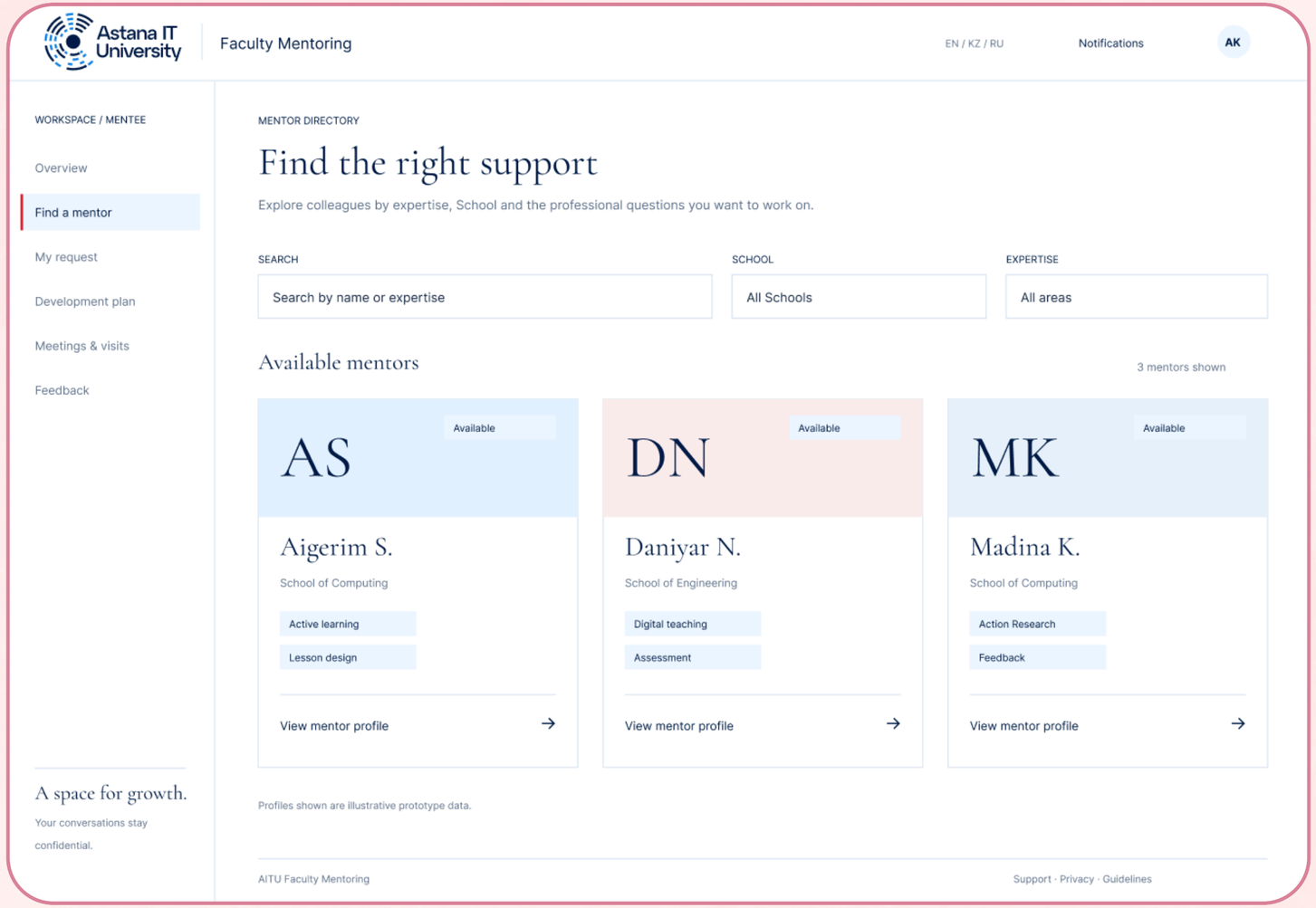


Figure 6. Mentor catalogue and selection.

MENTEE REQUEST FORM

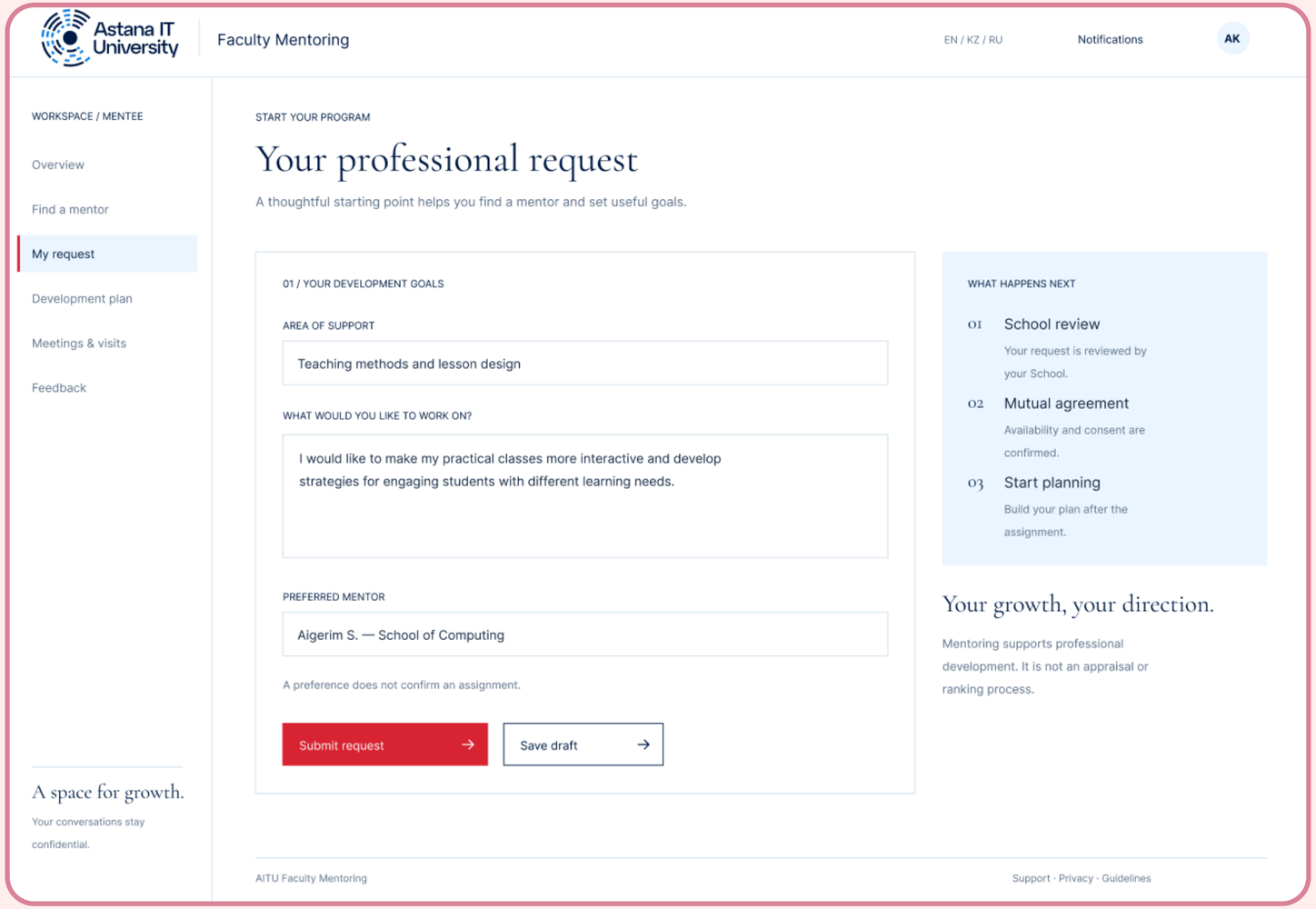


Figure 7. Mentee request form.

INDIVIDUAL DEVELOPMENT PLAN

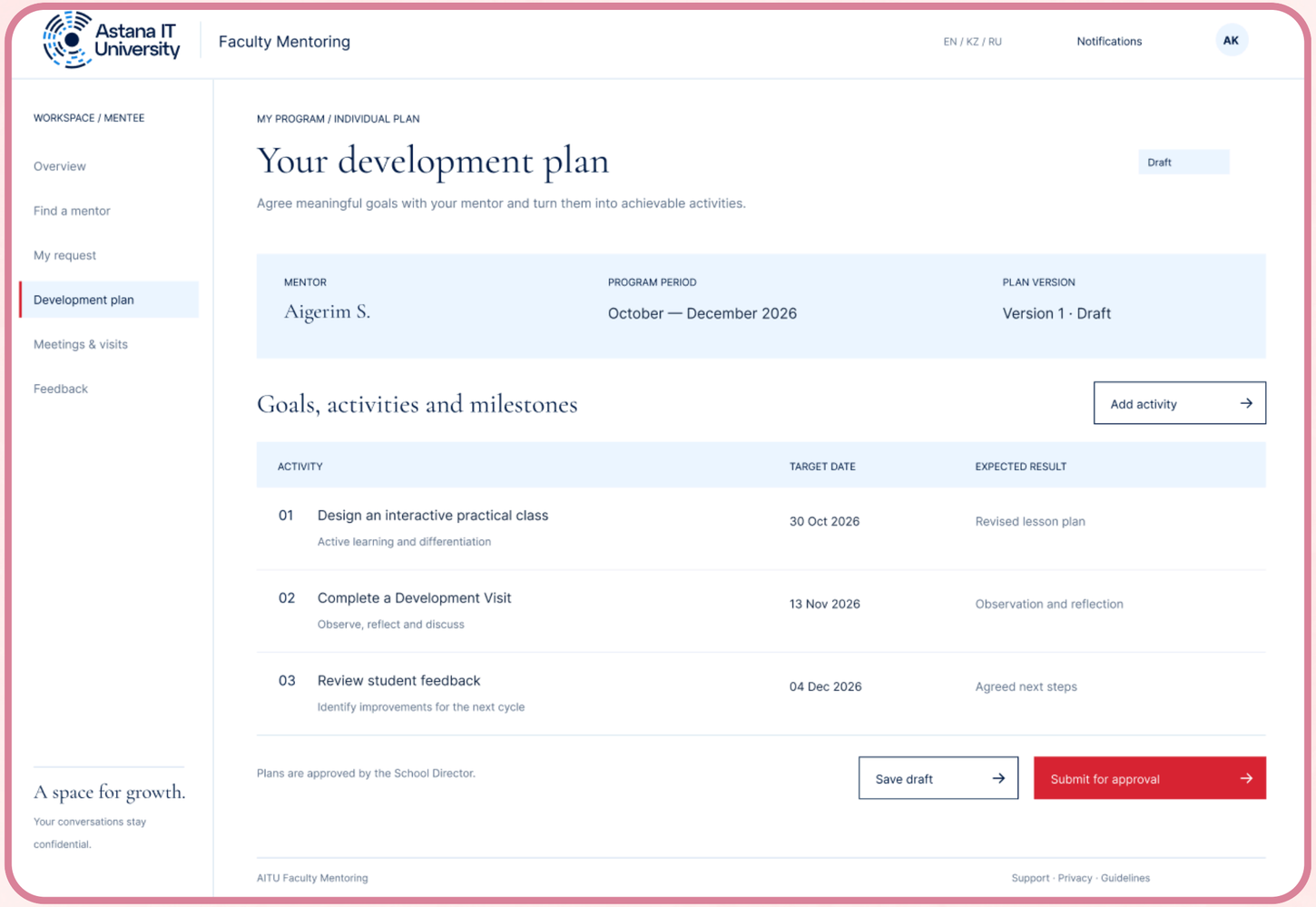


Figure 8. Individual development plan.

MENTOR APPLICATION FORM

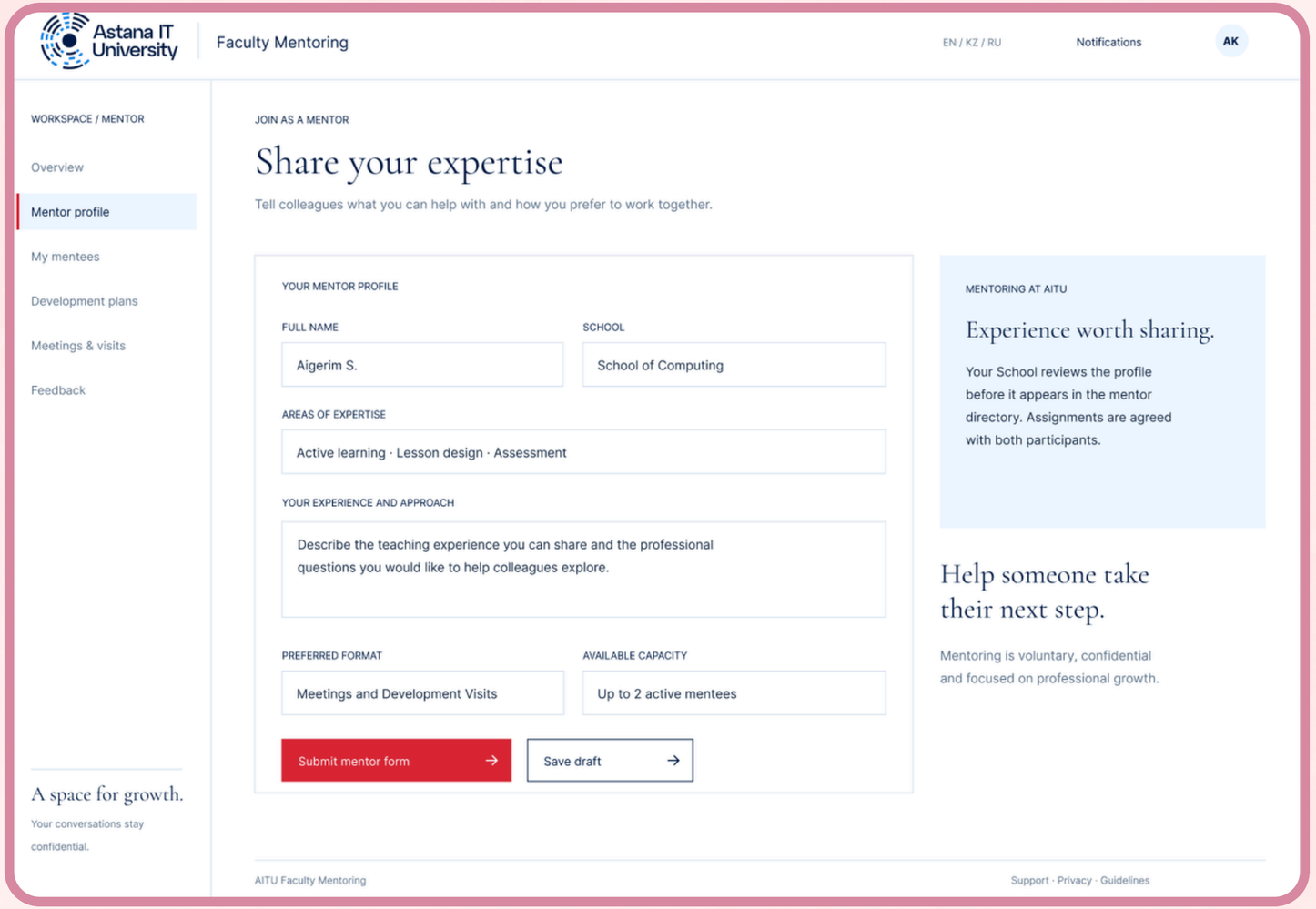


Figure 9. Mentor profile and application form.

MENTORING EVENTS

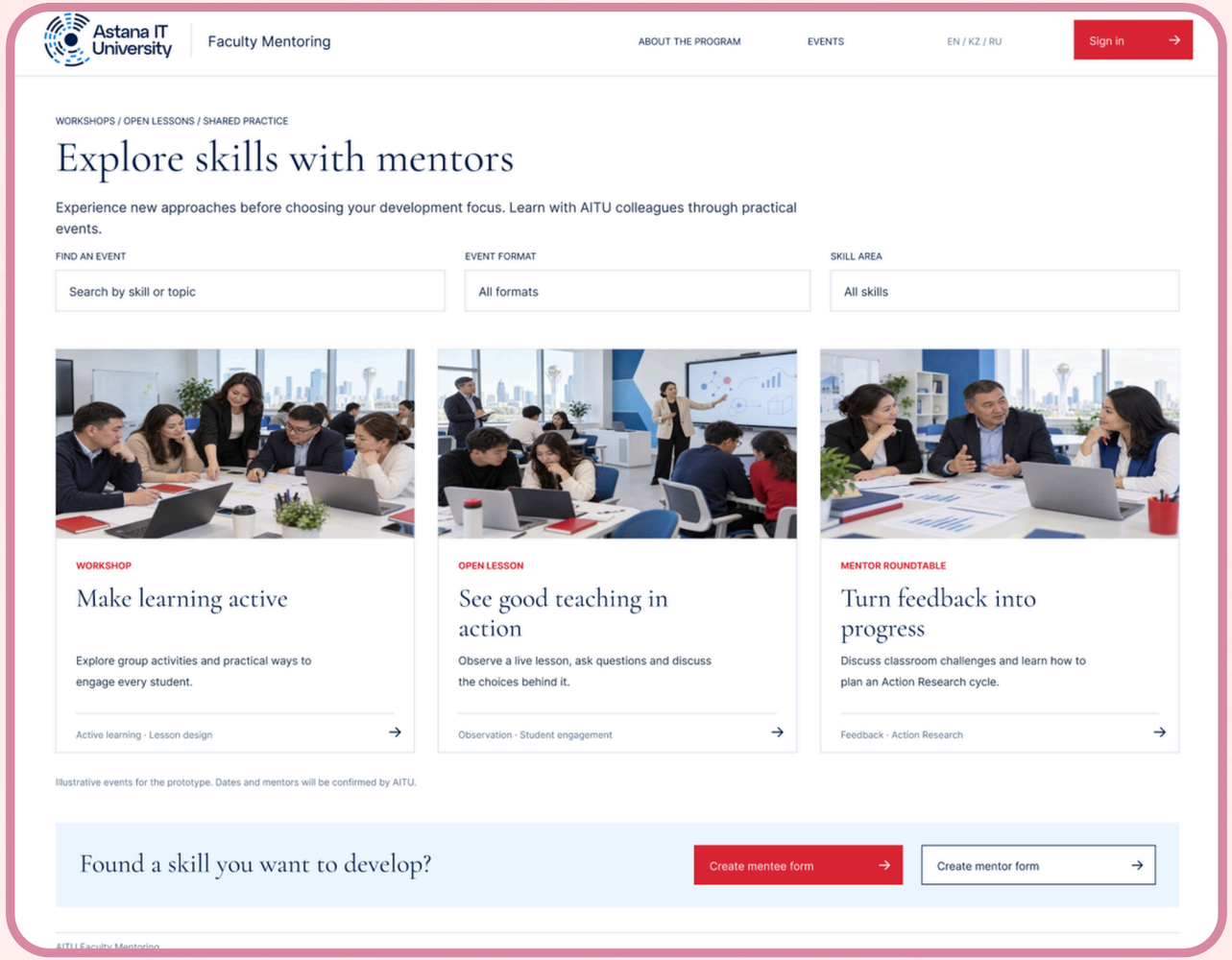


Figure 10. Workshops and open lessons catalogue.

VERIFICATION AND EVALUATION PLAN

Area	Testing method and conditions	Evaluation metrics and targets
Confidentiality	Attempt access across different roles and mentoring programs through the interface and API.	Zero unauthorised access attempts succeed in the defined test cases.
Availability under high load	Gradually increase concurrent users performing sign-in, form submission and search in a fixed test environment.	Measure successful request rate, response time and failures.
Corporate Microsoft authentication	Test valid AITU Microsoft accounts, unauthorised accounts and expired sessions through Microsoft Entra ID.	All defined authentication tests pass: authorised users sign in, unauthorised accounts are rejected, and expired sessions require reauthentication.

Table 13. Verification and evaluation plan.

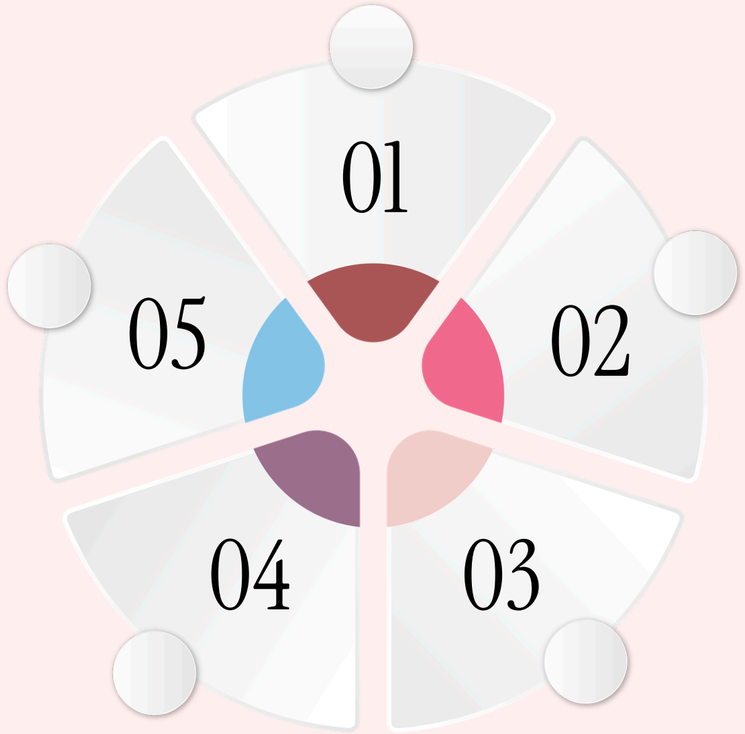
SCOPE AND DELIVERY PLAN

Core platform scope

Requests and mentor profiles; School-led appointments; approved plans; meetings; Development Visits; final forms and archive.

Later / conditional scope

Corporate SSO, HR and signing integrations depend on access. ML recommendations require suitable real data and local validation.



Validate design

Confirm requirements, permissions and prototype with client.

Pilot and evaluate

Test acceptance scenarios, compare baseline and report limits.

Implement core

Build access control and one complete mentoring lifecycle.

PROJECT MANAGEMENT AND RISKS

Team responsibilities	Project timeline	Project risks (P / I)	Mitigation strategy
Karina Tulegenova Planning, frontend and backend development, testing and documentation.	Sep–Oct 2026: requirements and literature review. Nov 2026: design and architecture.	Development delays Medium / High	Prioritise the core mentoring lifecycle. Review milestones weekly and keep a time buffer.
Yersultan Tulebayev Supervisor: review scope, architecture and evaluation.	Dec 2026–Feb 2027: core platform development. Mar 2027: integration and system testing.	Integration or access delays — Medium probability / High impact.	Prototype interfaces early. Use mock services while SSO / HR access is pending.
AITU stakeholders Confirm workflow and permissions, review prototypes and join the pilot.	Apr 2027: pilot and evaluation. May 2027: final fixes, thesis and defence.	Sensitive feedback exposure Low / High	Apply role permissions and access tests. Use anonymised pilot data and restrict feedback visibility.

Planned tools: GitHub for code and reviews. YouTrack for tasks and milestones. Project URLs: to be confirmed.

SUPERVISOR FEEDBACK AND IMPROVEMENTS

#	Supervisor's feedback	Action taken	Status
1	Use full publication titles and consolidate Literature Review.	Added complete titles for all 12 studies. Combined four review tables into one thematic overview.	Completed
2	Add Project Management and Risks. Extend the timeline from September 2026 to May 2027.	Added responsibilities, a nine-month delivery plan and project risks with mitigation actions.	Completed
3	Add Supervisor Feedback and Improvements before References.	Added a traceable feedback table and an example of the literature review improvement.	Completed

Before: four review slides with shortened study labels. After: one thematic table with 12 full publication titles and findings.

REFERENCES

[1] W. Nuis, M. Segers, and S. Beausaert, "Conceptualizing mentoring in higher education: A systematic literature review," Educ. Res. Rev., vol. 41, Art. no. 100565, 2023, doi: 10.1016/j.edurev.2023.100565.

[2] M. Abdollahi and F. Heshmati Nabavi, "Mentoring as an appropriate strategy for medical faculty member development in higher education: A systematic review," J. Adv. Med. Educ. Prof., vol. 11, no. 1, pp. 3–14, 2023, doi: 10.30476/JAMP.2022.97103.1740.

[3] M. López-Aguado and O. Scialdone, "La mentoría como estrategia para la mejora docente en la universidad, una revisión sistemática," Rev. Complut. Educ., vol. 37, no. 1, pp. 21–33, 2026, doi: 10.5209/rced.96870.

[4] K. Bell, "Affordances, constraints and collaborative potential of e-mentoring platforms and their features for K–12 educators: A systematic literature review," Int. J. Mentor. Coach. Educ., vol. 13, no. 3, pp. 346–360, 2024, doi: 10.1108/IJMCE-07-2023-0056.

[5] F. Jin, Z. Song, W. M. Cheung, C.-H. Lin, and T. Liu, "Technological affordances in teachers' online professional learning communities: A systematic review," J. Comput. Assist. Learn., vol. 40, no. 3, pp. 1019–1039, 2024, doi: 10.1111/jcal.12935.

[6] L. Huang, M. Liang, Y. Xiong, X. Wu, and C. P. Lim, "A systematic review of technology-enabled teacher professional development during COVID-19 pandemic," Comput. Educ., vol. 223, Art. no. 105168, 2024, doi: 10.1016/j.compedu.2024.105168.

[7] L. Cutroni and A. Paladino, "Peer-ing in: A systematic review and framework of peer review of teaching in higher education," Teach. Teach. Educ., vol. 133, Art. no. 104302, 2023, doi: 10.1016/j.tate.2023.104302.

[8] F. M. Otero-Saborido, J. A. Domínguez-Montes, J. M. Cenizo-Benjumea, and G. González-Calvo, "Peer observation of teaching in higher education: Systematic review of observation tools," Educ. Process: Int. J., vol. 13, no. 1, pp. 84–101, 2024, doi: 10.22521/edupij.2024.131.6.

[9] B. De la Iglesia, D. Forteza, and L. Duma, "Peer feedback in teacher professional development: A systematic review," Rev. Esp. Pedagog., vol. 82, no. 288, pp. 335–358, 2024, doi: 10.22550/2174-0909.4044.

[10] E. Cano García, "Peer feedback for teaching professional development: Conditions for it to take effect," Cogent Educ., vol. 11, no. 1, Art. no. 2391577, 2024, doi: 10.1080/2331186X.2024.2391577.

[11] P. Hunt, Ä. Leijen, and M. van der Schaaf, "Automated feedback is nice and human presence makes it better: Teachers' perceptions of feedback by means of an e-portfolio enhanced with learning analytics," Educ. Sci., vol. 11, no. 6, Art. no. 278, 2021, doi: 10.3390/educsci11060278.

[12] A. Wang, S. Yu, M. Wang, and L. Chen, "Effects of the visual analytics of peer feedback on teachers' TPACK development in a lesson study," Interact. Learn. Environ., vol. 32, no. 8, pp. 4632–4651, 2024, doi: 10.1080/10494820.2023.2204354.

[13] Together, "Together mentorship platform." Accessed: Oct. 15, 2026. [Online]. Available: https://www.togetherplatform.com/page/mentorship-software-platform

[14] Chronus, "Mentoring software." Accessed: Oct. 15, 2026. [Online]. Available: https://chronus.com/software/mentoring-software

[15] IRIS Connect, "Video technology." Accessed: Oct. 15, 2026. [Online]. Available: https://www.irisconnect.com/us/video-technology/

[16] IRIS Connect, "Instructional coaching." Accessed: Oct. 15, 2026. [Online]. Available: https://www.irisconnect.com/uk/instructional-coaching/

[17] IRIS Connect, "Progress tracking (Objectives)." Accessed: Oct. 15, 2026. [Online]. Available: https://www.irisconnect.com/uk/progress-tracking/

[18] IRIS Connect, "Adaptive professional development." Accessed: Oct. 15, 2026. [Online]. Available: https://www.irisconnect.com/uk/adaptive-pd/

[19] Chronus, "Mentor matching." Accessed: Oct. 15, 2026. [Online]. Available: https://chronus.com/software/mentoring-software/mentor-matching

[20] Chronus, "Privacy Notice." Accessed: Oct. 15, 2026. [Online]. Available: https://mentoring365.chronus.com/p/pl/privacy_policy